CAFE-Fusion v2: Complete End-to-End Multimodal Marine Oil Spill Architecture

SAR Modality

Single-Look Complex (SLC)
Radar Backscatter
• Day/Night & All-Weather
• Sensitive to Capillary Damping

Frozen CBD-Net
(Unimodal SAR Expert)

F1: 0.8358 | IoU: 0.7199

Hyperspectral (HSI)

128 Spectral Bands
Optical Reflectance Cube
• High Diagnostic Hydrocarbon Bands
• Dominant Global Sensor

Frozen HMOSDN
(Unimodal HSI Expert)

F1: 0.9261 | IoU: 0.8636

11-D Gate Vector (u)

```
u0: p_SAR (SAR Probability)
u1: z_SAR (SAR Logit)
u2: p_HSI (HSI Probability)
u3: z_HSI (HSI Logit)
u4: |p_SAR - p_HSI| (Abs Diff)
u5: p_SAR - p_HSI (Signed Diff)
u6: c_HSI (HSI Confidence)
u7: H(p_HSI) (HSI Entropy)
u8: H(p_SAR) (SAR Entropy)
u9: μ_SAR (Local SAR Mean)
u10: σ_SAR (Local SAR Std Dev)
```

Tiny Reliability Gate

```
Input: u in R^11
  -> Linear(11 -> 64) + LayerNorm
  -> GELU + Dropout(0.15)
  -> Linear(64 -> 32) + LayerNorm
  -> GELU
  -> Linear(32 -> 1)
Bias Init: b3 = -2.0
g_init = sigmoid(-2.0) approx 0.119 (HSI Anchor)
```

Output: Dynamic Gate g in [0, 1]

Masked Reliability Loss

```
m_i = c_SAR XOR c_HSI (Conflict Mask)
If HSI Right & SAR Wrong -> y_rel = 0
If SAR Right & HSI Wrong -> y_rel = 1
L_rel = BCE_masked(g, y_rel)
Protects HSI & Trains Conflict Routing
```

Convex Logit Fusion

```
z_fusion = (1-g)*z_HSI + g*z_SAR

p_fusion = sigmoid(z_fusion)

Decision: y_pred = I(p >= t*)
```

```
Total Loss:
L = L_BCE + L_Dice
  + 0.10*L_rel + 0.01*mean(g)
```

```
6-Fold Primary Result:
Mean F1:  0.9288 +/- 0.0270
Mean IoU: 0.8679 +/- 0.0465
Beats HSI on 5 of 6 folds
Recovers 11.34% Oracle Gap
```